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10. 3차원 공간

입체 공간의 생성

Heesung Oh
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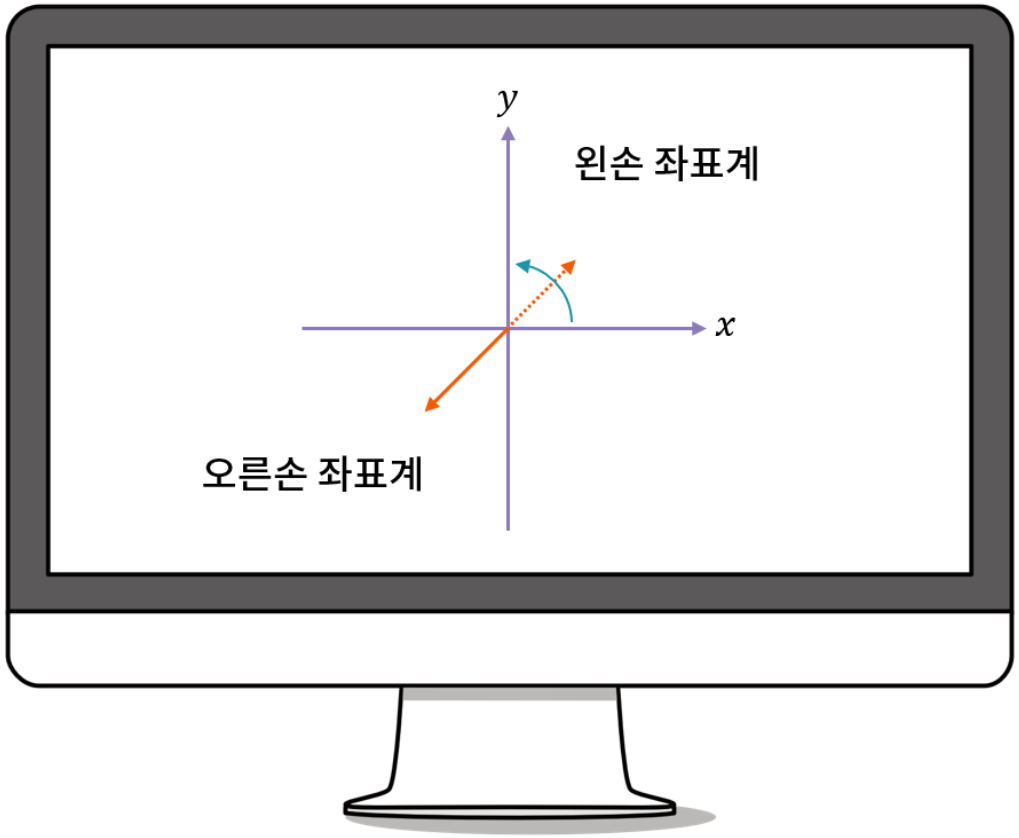
- **오른손 좌표계 vs 왼손 좌표계**



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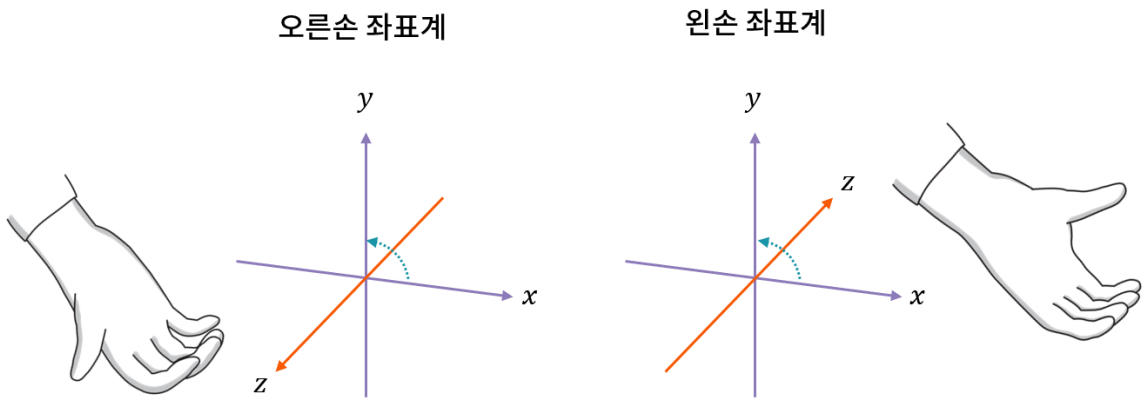
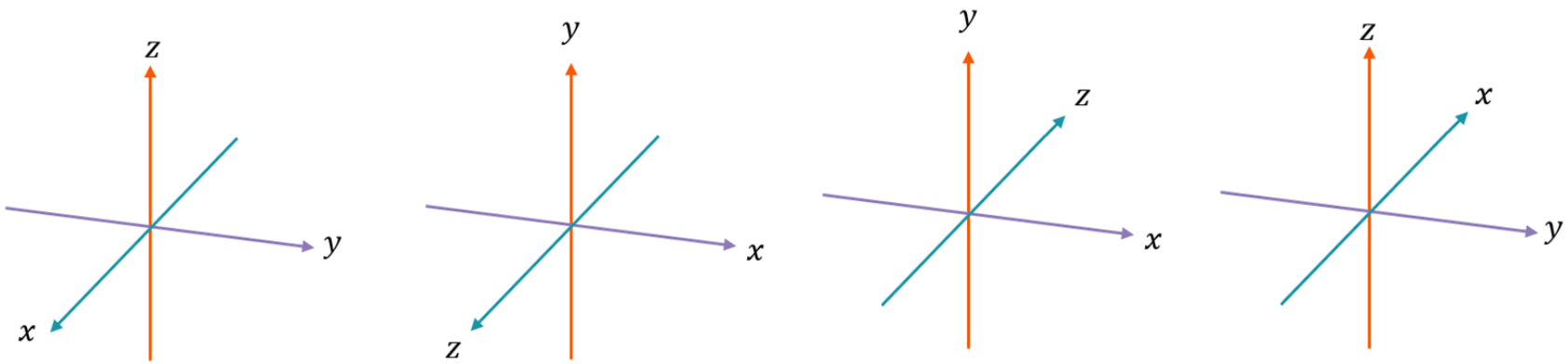
- 오른손 좌표계 왼손 좌표계



- 3D Max, OpenGL, Unity, Unreal



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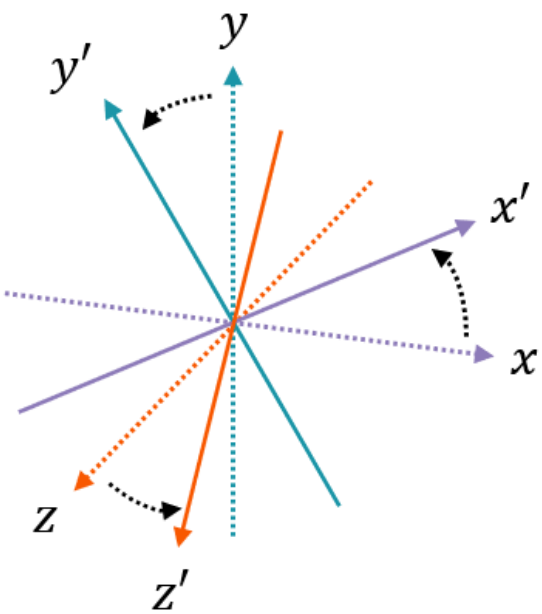
- **세 축의 기저벡터로 만들어지는 회전행렬**

$$S = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad T = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$R = \begin{bmatrix} x' & y' & z' \end{bmatrix}$$

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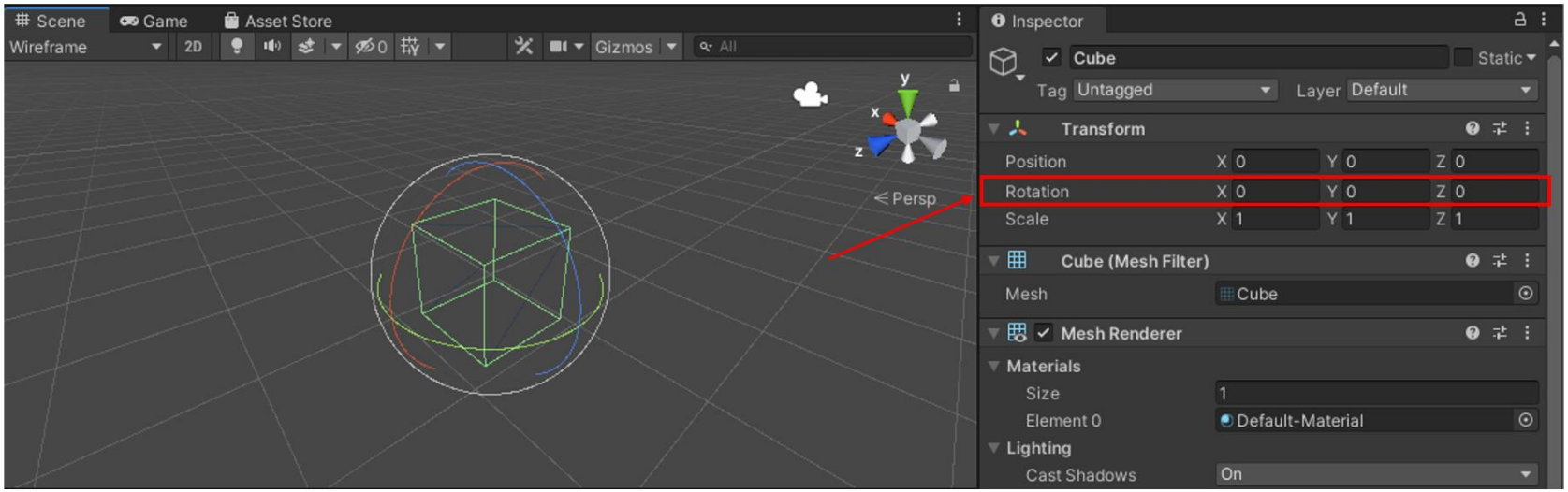
- **오일러 회전**
  - 3차원 좌표계(x, y, z)에서 X, Y, Z 축을 기준으로 물체를 순차적으로 회전, 방향을 결정하는 방식

- **유니티 엔진 오일러 각 3차원 회전**



$$(\theta_x, \theta_y, \theta_z)$$

10.2.1 오일러 각

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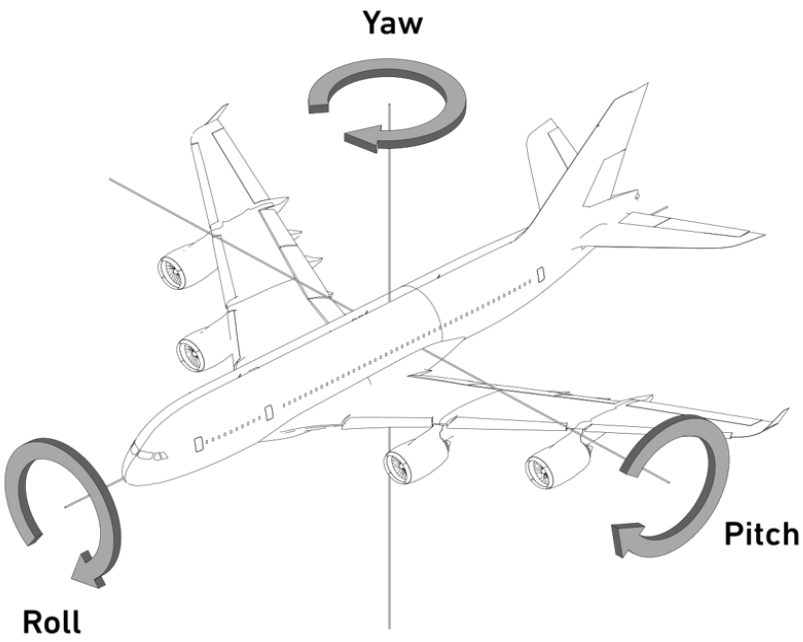
- 오일러 각: vector 표현

$$(\theta_x, \theta_y, \theta_z)$$

- 요(Y: Yaw), 롤(Z: Roll), 피치(X: Pitch) 회전
  - 표준 기저 벡터 축을 기준으로 회전 ← 항공기 회전 표현, 게임에서 사용



10.2.1 오일러 각

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- X 축 회전

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

- Y 축 회전

$$R_y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

- Z 축 회전

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- 축 회전: X → Y → Z → X → Y
  - Y축에 직교하는 평면: zx (0) xz(X)
  - X축에 직교하는 평면: yz
  - Z축에 직교하는 평면: xy

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- **오일러 각 조합에 대한 경우의 수**

경우	회전축의 순서	회전 동작의 순서
1	x → y → z	피치 → 요 → 롤
2	x → z → y	피치 → 롤 → 요
3	y → x → z	요 → 피치 → 롤
4	y → z → x	요 → 롤 → 피치
6	**z → y → x**	**롤 → 요 → 피치**

- $\boldsymbol{R = R_{yaw} \cdot R_{pitch} \cdot R_{roll}}$

$$R = R_\alpha \cdot R_\beta \cdot R_\gamma$$

$$= \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\beta & -\sin\beta \\ 0 & \sin\beta & \cos\beta \end{bmatrix} \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\alpha\cos\gamma + \sin\alpha\sin\beta\sin\gamma & -\cos\alpha\sin\gamma + \sin\alpha\sin\beta\cos\gamma & \sin\alpha\cos\beta \\ \cos\beta\sin\gamma & \cos\beta\cos\gamma & -\sin\beta \\ -\sin\theta\cos\gamma + \cos\alpha\sin\beta\sin\gamma & \sin\alpha\sin\gamma + \cos\alpha\sin\beta\cos\gamma & \cos\alpha\cos\beta \end{bmatrix}$$

10.2.2 회전 행렬의 유도

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- X, Y, Z

$$X_{local} = (\cos\alpha\cos\gamma + \sin\alpha\sin\beta\sin\gamma\,,\ \cos\beta\sin\gamma\,,\ -\sin\alpha\cos\gamma + \cos\alpha\sin\beta\sin\gamma)$$

$$Y_{local} = (-\cos\alpha\sin\gamma + \sin\alpha\sin\beta\cos\gamma\,,\ \cos\beta\cos\gamma\,,\ \sin\alpha\sin\gamma + \cos\alpha\sin\beta\cos\gamma)$$

$$Z_{local} = (\sin\alpha\cos\beta\,,\ -\sin\beta\,,\ \cos\alpha\cos\beta)$$

- 로컬 축 벡터와 회전 행렬
  - 로컬 축 벡터

$$\vec{x} = (x_x, x_y, x_z) \quad \vec{y} = (y_x, y_y, y_z) \quad \vec{z} = (z_x, z_y, z_z)$$

  - 회전 행렬

$$R_{RHL} = \begin{bmatrix} x_x & y_x & z_x & 0 \\ x_y & y_y & z_y & 0 \\ x_z & y_z & z_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

10.2.3 3차원 모델링 행렬

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- 크기(S), 회전(R), 이동(T)

$$S = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_{RHL} = \begin{bmatrix} x_x & y_x & z_x & 0 \\ x_y & y_y & z_y & 0 \\ x_z & y_z & z_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- T·R·S 모델링행렬 M

$$M = T \cdot R \cdot S = \begin{bmatrix} x_x s_x & y_x s_y & z_x s_z & t_x \\ x_y s_x & y_y s_y & z_y s_z & t_y \\ x_z s_x & y_z s_y & z_z s_z & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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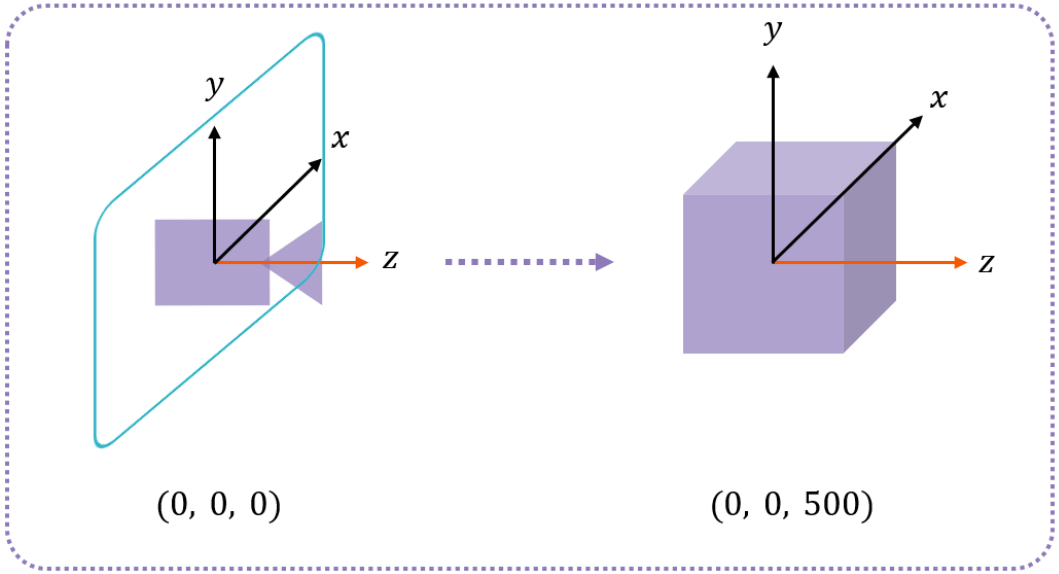
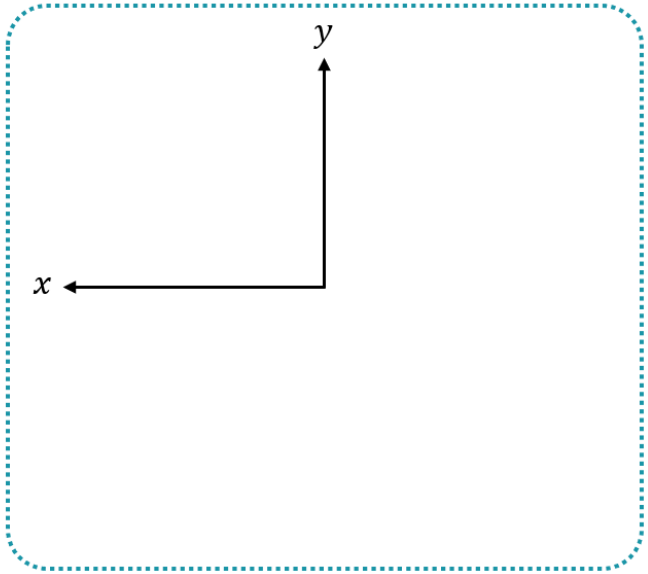
3차원 공간의 카메라 설정

월드 공간



카메라가 만드는 화면 좌표계



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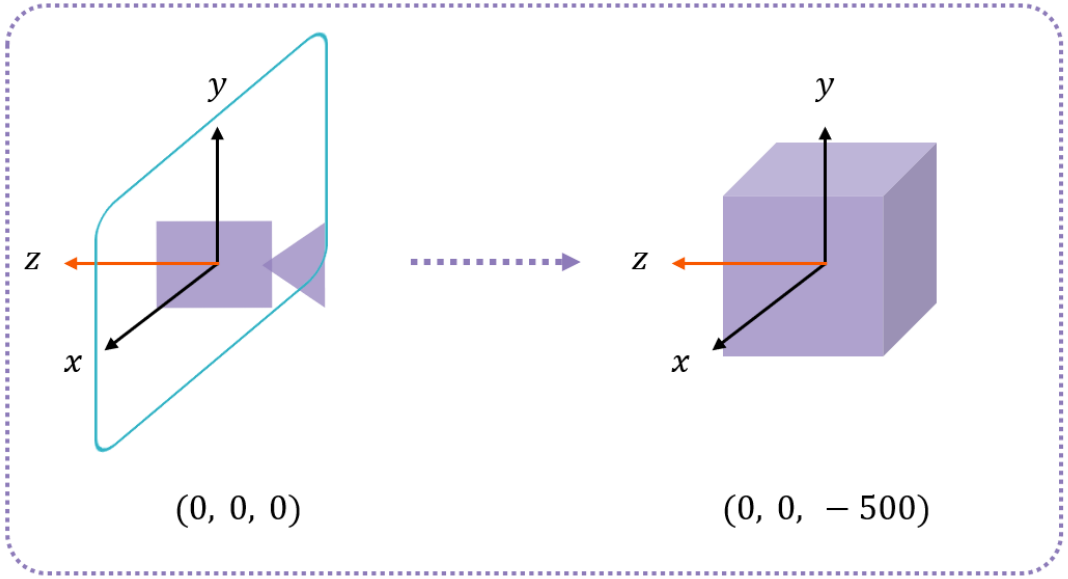
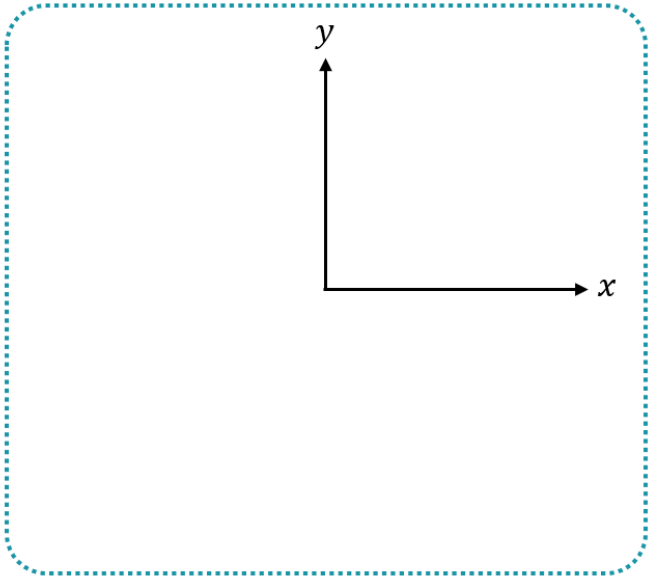
뷰 공간으로 재해석된 3차원 공간

뷰 공간



카메라가 만드는 화면 좌표계



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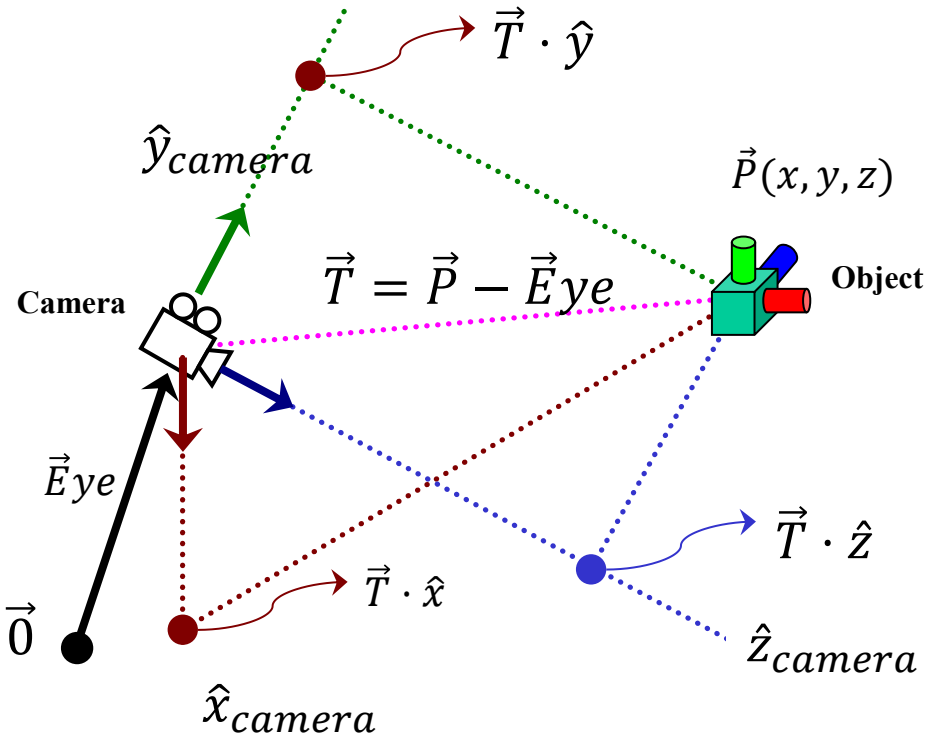
- 뷰 공간으로 재해석된 3차원 공간



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- **뷰 행렬**
  - **카메라 지역 공간으로 이동**
    - 카메라 이동의 역행렬 → 회전의 역행렬 적용
    - 크기 변환 없음. 회전, 이동 변환 만 존재
  - **카메라 축:** $\hat{x}, \hat{y}, \hat{z}$, **카메라 transform :** $\vec{t} = (t_x, t_y, t_z)$

- **M_view = R_camera $^{-1}$ · T_camera $^{-1}$**

$$R^{-1} = \begin{bmatrix} x_x & x_y & x_z & 0 \\ y_x & y_y & y_z & 0 \\ z_x & z_y & z_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T = \begin{bmatrix} 1 & 0 & 0 & -t_x \\ 0 & 1 & 0 & -t_y \\ 0 & 0 & 1 & -t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$M_{view} = \begin{bmatrix} x_x & x_y & x_z & -\hat{x} \cdot \vec{t} \\ y_x & y_y & y_z & -\hat{y} \cdot \vec{t} \\ z_x & z_y & z_z & -\hat{z} \cdot \vec{t} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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- Y 축으로 180도 회전

$$V = \begin{bmatrix} -x_x & -x_y & -x_z & \hat{x} \cdot \vec{t} \\ y_x & y_y & y_z & -\hat{y} \cdot \vec{t} \\ -z_x & z_y & -z_z & +\hat{z} \cdot \vec{t} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

10-1 처음 그려보는 3차원 공간

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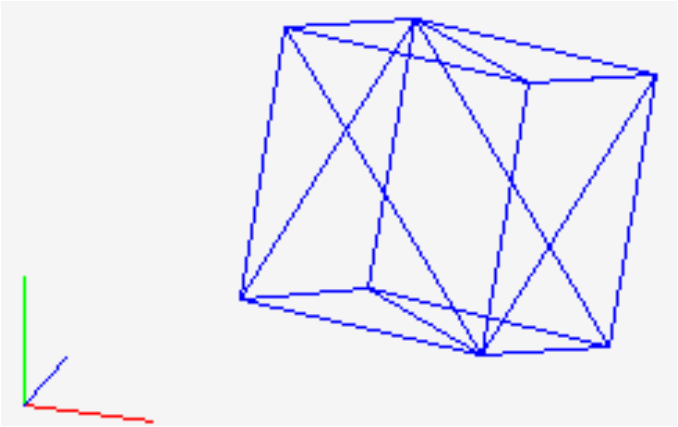
```
FORCEINLINE void CameraObject::GetViewAxes(Vector3& OutViewX, Ve
ctor3& OutViewY, Vector3& OutViewZ) const
{
  // 로컬 Z축 값의 반대 방향을 뷰 공간의 Z축으로 설정
  OutViewZ = -_Transform.GetLocalZ();
  OutViewY = _Transform.GetLocalY();
  OutViewX = -_Transform.GetLocalX();
}
```



```
FORCEINLINE Matrix4x4 CameraObject::GetViewMatrix() const
{
  Vector3 viewX, viewY, viewZ;
  GetViewAxes(viewX, viewY, viewZ);
  Vector3 pos = _Transform.GetPosition();

  return Matrix4x4(
    Vector4(Vector3(viewX.X, viewY.X, viewZ.X), false),
    Vector4(Vector3(viewX.Y, viewY.Y, viewZ.Y), false),
    Vector4(Vector3(viewX.Z, viewY.Z, viewZ.Z), false),
    Vector4(-viewX.Dot(pos), -viewY.Dot(pos), -
viewZ.Dot(pos), 1.f)
  );
}
```

10.4 오일러 각의 특징

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- 장점
  - 회전에 대한 직관적인 인터페이스 제공
  - 3개의 데이터 만 있으면 회전을 표현
    (회전을 표현하려면 3x3 행렬원소 필요)
  - 직관적인 표준 기저 벡터 를 회전축으로 사용 → 회전 설계 용이
  - 회전 설정에서 가장 적합 → 대다수 3차원 소프트웨어에서 사용

- 단점
  - 짐벌락 현상

10.4.1 짐벌락 현상

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- 짐벌락 현상
  - 특정 각도에서 두 개의 회전축이 겹쳐, 하나의 자유도 상실
- 발생 원인
  - 행렬 연산 순서의 종속성
- 해결 방법
  - 쿼터니언(Quaternion) 사용

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- **오일러 각도로 인한 자유도 손실**
  https://en.wikipedia.org/wiki/Gimbal_lock

$$R = R_\alpha \cdot R_\beta \cdot R_\gamma$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **$\beta = \pi/2$**

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ \sin\alpha\cos\gamma + \cos\alpha\sin\gamma & -\sin\alpha\sin\gamma + \cos\alpha\cos\gamma & 0 \\ -\cos\alpha\cos\gamma + \sin\alpha\sin\gamma & \cos\alpha\sin\gamma + \sin\alpha\cos\gamma & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ \sin(\alpha+\gamma) & \cos(\alpha+\gamma) & 0 \\ -\cos(\alpha+\gamma) & \sin(\alpha+\gamma) & 0 \end{bmatrix}$$

$$R = f(\alpha+\gamma) = f\big((\alpha+\Delta)+\gamma\big) = f\big(\alpha+(\gamma+\Delta)\big)$$

  - **α와 γ의 회전 효과를 구분할 수 없음**

- **수학적: 회전 행렬이 α와 γ를 독립 변수로 가지지 않고 $\alpha + \gamma$로만 표현**
- **물리적: 두 회전 축이 정렬되어 자유도 1개가 소실**

10-2 짐벌락 현상의 체험

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```
// 입력에 따른 플레이어 트랜스폼의 변경
Rotator r = goPlayer.GetTransform().GetRotation();
if (input.IsPressed(InputButton::Space))
{
  r.Pitch = -90.f;
}
else
{
  r.Pitch += input.GetAxis(InputAxis::ZAxis) * rotateSpeed * InDeltaSeconds;
}
r.Roll += input.GetAxis(InputAxis::XAxis) * rotateSpeed * InDeltaSeconds;
r.Yaw += input.GetAxis(InputAxis::YAxis) * rotateSpeed * InDeltaSeconds;
```

10.4.2 회전 보간의 계산

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- **회전의 선형 보간**

$$\theta' = (1-t)\,\theta_{start} + t\,\theta_{end}$$

- **특정 축 회전** $R(\alpha, 0,0)$

$$R(\alpha,\theta,\varphi) = R_\alpha \cdot R_\theta \cdot R_\varphi$$
$$R(\beta,\theta,\varphi) = R_\beta \cdot R_\theta \cdot R_\varphi$$
$$R(\beta,\theta,\varphi) \cdot R(\alpha,\theta,\varphi) = R_\beta \cdot R_\theta \cdot R_\varphi \cdot R_\alpha \cdot R_\theta \cdot R_\varphi \quad = R_\beta \cdot I \cdot I \cdot R_\alpha \cdot I \cdot I$$
$$R(\beta,\theta,\varphi) \cdot R(\alpha,\theta,\varphi) = R_\beta \cdot R_\alpha = R_{(\alpha+\beta)}$$

- **특정 축 회전** $R(\alpha, \alpha, 0)$

$$R(\beta) \cdot R(\alpha) \neq R_{(\alpha+\beta)}$$

10.5 정리

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- 3차원 트랜스폼
  - T·R·S

- 카메라 뷰 행렬

$$M_{view} = \begin{bmatrix} x_x & x_y & x_z & -\hat{x} \cdot \vec{t} \\ y_x & y_y & y_z & -\hat{y} \cdot \vec{t} \\ z_x & z_y & z_z & -\hat{z} \cdot \vec{t} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- 오일러 각
  - 짐벌락
  - 사원수 보간

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11. 외적

3차원 공간의 분석과 응용

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- 외적
  - Cross product, Outer product

$\vec{u} = (u_x, u_y, u_z) \quad \vec{v} = (v_x, v_y, v_z)$

$$\vec{u} \times \vec{v} = |\vec{u}||\vec{v}| \sin\theta \ \hat{n}$$

$$\vec{u} \times \vec{v} = \varepsilon_{ijk} u_i v_j \hat{e}_k$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$$

$$\vec{u} \times \vec{v} = (u_y v_z - u_z v_y,\ u_z v_x - u_x v_z,\ u_x v_y - u_y v_x)$$

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- **외적의 성질**

$$\vec{v} \times \vec{u} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ u_x & u_y & u_z \end{vmatrix} \begin{aligned} &= (v_y u_z - v_z u_y,\ v_z u_x - v_x u_z,\ v_x u_y - u_y v_x) \\ &= -(v_z u_y - v_y u_z,\ v_x u_z - v_z u_x,\ u_y v_x - v_x u_y) \\ &= \ -\vec{u} \times \vec{v} \end{aligned}$$

$$\vec{u} \times \vec{v} \ = \ -\vec{v} \times \vec{u}$$

- **외적 덧셈 분배 법칙**

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} \ + \vec{A} \times \vec{C}$$

- **3항 연산**

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

$$(\vec{A} \times \vec{B}) \times \vec{C} = -\vec{C} \times (\vec{A} \times \vec{B}) = -(\vec{C} \cdot \vec{B})\vec{A} + (\vec{C} \cdot \vec{A})\vec{B} = (\vec{A} \cdot \vec{C})\vec{B} \ - (\vec{B} \cdot \vec{C})\vec{A}$$

$$\vec{A} \times (\vec{B} \times \vec{C}) \ \neq (\vec{A} \times \vec{B}) \times \vec{C}$$

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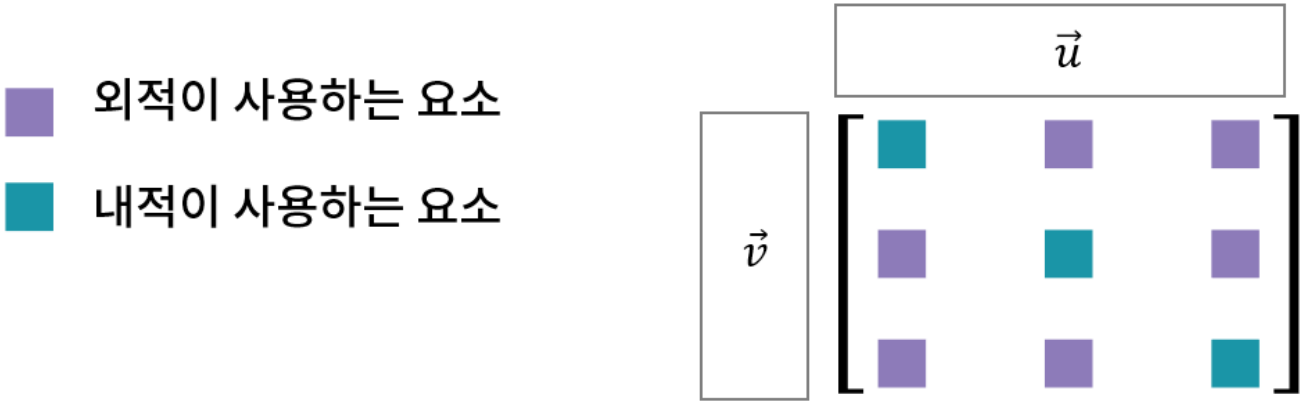
- **내적과 외적이 가지는 성질 비교**



- **벡터 내적 외적 차이**

	내적	외적
계산 결과	**스칼라**	**벡터**
교환 법칙	성립함	성립하지 않음
결합 법칙	성립하지 않음	성립하지 않음
분배 법칙	성립함	성립함
연산 방법	같은 위치의 요소만 사용	다른 위치의 요소만 사용

11.1.1 평행성 판별

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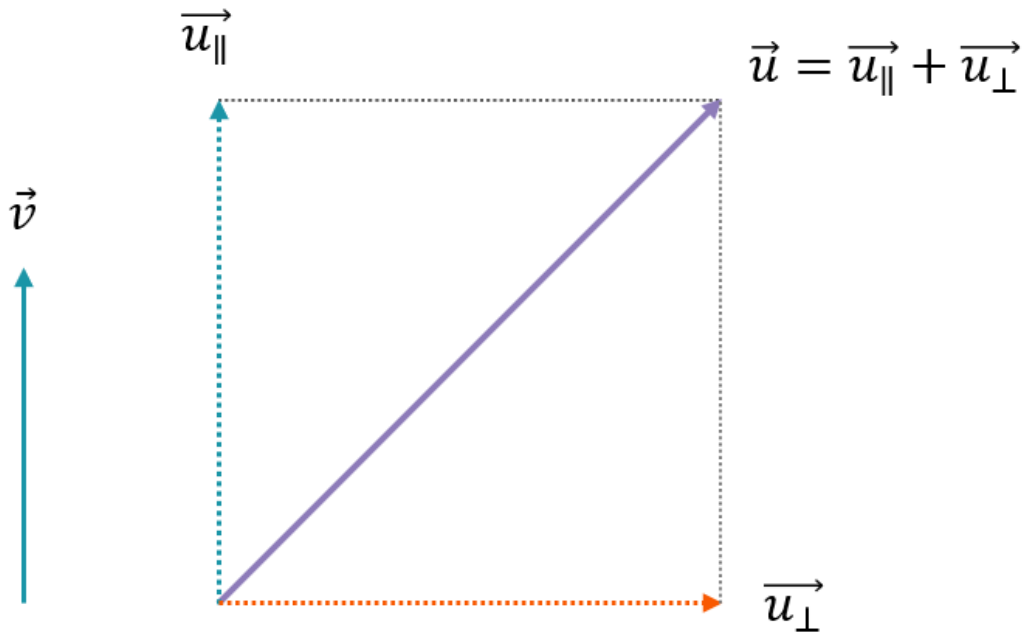
- 두 벡터의 수직: 내적 결과 0
- 두 벡터의 수평: 외적 결과 0-벡터
- 벡터를 수평 성분, 수직 성분 분리



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- $\vec{u} \times \vec{v} = |\vec{u}||\vec{v}| sin\theta\, \hat{n}$ **전개**

$$|\vec{u} \times \vec{v}|^2 = |\vec{u}|^2|\vec{v}|^2 \sin^2\theta$$

$$= |\vec{u}|^2|\vec{v}|^2(1 - \cos^2\theta)$$

$$= |\vec{u}|^2|\vec{v}|^2 - |\vec{u}|^2|\vec{v}|^2\cos^2\theta$$

$$= |\vec{u}|^2|\vec{v}|^2 - (\vec{u}\cdot\vec{v})^2$$

$$= \left(u_x u_x + u_y u_y + u_z u_z\right)\left(v_x v_x + v_y v_y + v_z v_z\right) - \left(u_x v_x + u_y v_y + u_z v_z\right)^2$$

$$\begin{aligned}
&= u_x u_x v_x v_x + u_y u_y v_x v_x + u_z u_z v_x v_x \\
&+ u_x u_x v_y v_y + u_y u_y v_y v_y + u_z u_z v_y v_y \\
&+ u_x u_x v_z v_z + u_y u_y v_z v_z + u_z u_z v_z v_z \\
&- u_x v_x u_x v_x - u_y v_y u_y v_y - u_z v_z u_z v_z \\
&- 2u_x v_x u_y v_y - 2u_y v_y u_z v_z - 2u_z v_z u_x v_x
\end{aligned}$$

$$\begin{aligned}
&= u_y u_y v_x v_x + u_z u_z v_x v_x + u_x u_x v_y v_y + u_z u_z v_y v_y + u_x u_x v_z v_z + u_y u_y v_z v_z \\
&- 2u_x v_x u_y v_y - 2u_y v_y u_z v_z - 2u_z v_z u_x v_x
\end{aligned}$$

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- $\vec{u} \times \vec{v} = |\vec{u}||\vec{v}| sin\theta\, \hat{n}$ **전개**

$$|\vec{u} \times \vec{v}|^2 = |\vec{u}|^2|\vec{v}|^2 \sin^2\theta$$

$$= u_y u_y v_x v_x + u_z u_z v_x v_x + u_x u_x v_y v_y + u_z u_z v_y v_y + u_x u_x v_z v_z + u_y u_y v_z v_z$$
$$-2u_x v_x u_y v_y - 2u_y v_y u_z v_z - 2u_z v_z u_x v_x$$

$$= u_y^2 v_x^2 + u_z^2 v_x^2 + u_x^2 v_y^2 + u_z^2 v_y^2 + u_x^2 v_z^2 + u_y^2 v_z^2 - 2u_x v_x u_y v_y - 2u_y v_y u_z v_z - 2u_z v_z u_x v_x$$
$$= u_y^2 v_z^2 - 2u_y v_z u_z v_y + u_z^2 v_y^2 + u_z^2 v_x^2 - 2u_z v_x u_x v_z + u_x^2 v_z^2 + u_x^2 v_y^2 - 2u_x v_y u_y v_x + u_y^2 v_x^2$$

$$= \left(u_y v_z - u_z v_y\right)^2 + (u_z v_x - u_x v_z)^2 + \left(u_x v_y - u_y v_x\right)^2$$
$$= \left(u_y v_z - u_z v_y,\ u_z v_x - u_x v_z,\ u_x v_y - u_y v_x\right)^2$$

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}^2$$

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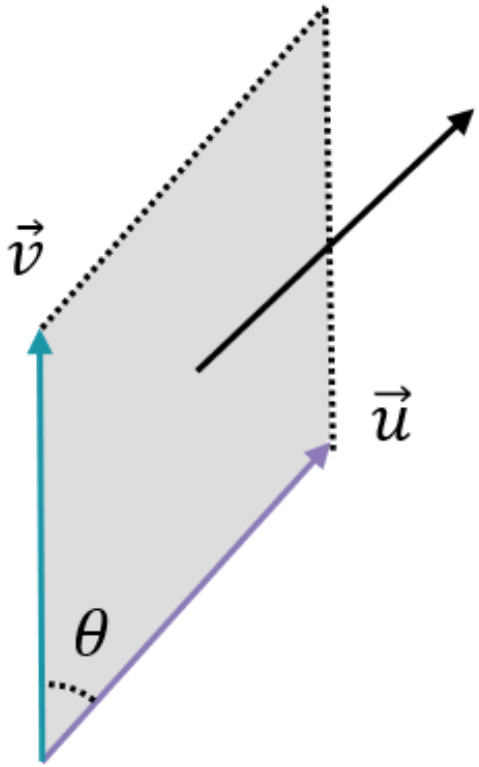
- 두 벡터 $\vec{u}, \vec{v}$ 가 만들어내는 평행사변형의 넓이



$$Area(P) = |\vec{u}||\vec{v}|\sin\theta$$

11.1.2 법선 벡터

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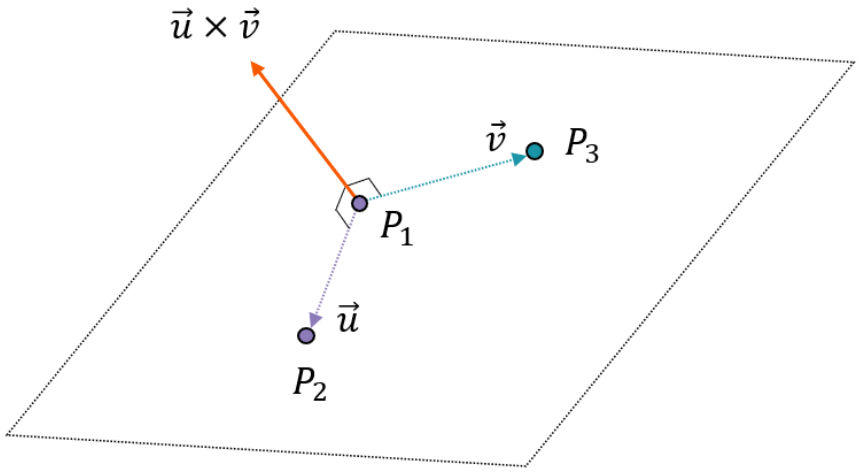
- **두 벡터의 외적 → 두 벡터에 직교**

$$\vec{u} \cdot \vec{u} \times \vec{v} = \begin{vmatrix} u_x & u_y & u_z \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = 0$$

$$\vec{v} \cdot \vec{u} \times \vec{v} = \begin{vmatrix} v_x & v_y & v_z \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = 0$$

- **법선 벡터를 생성하는 외적 연산**



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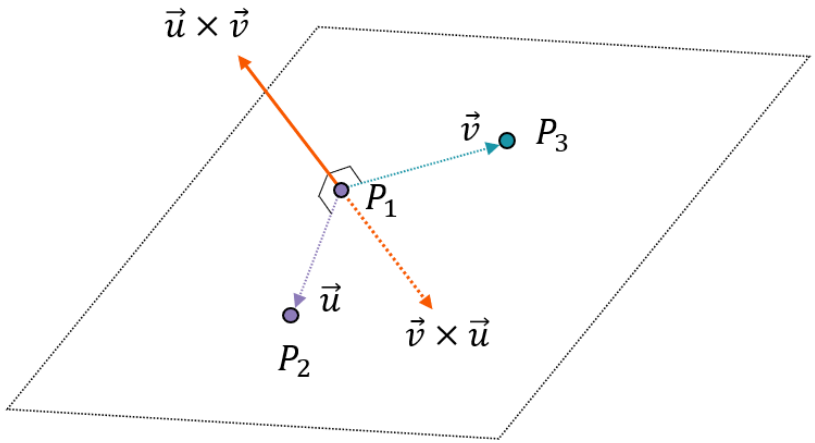
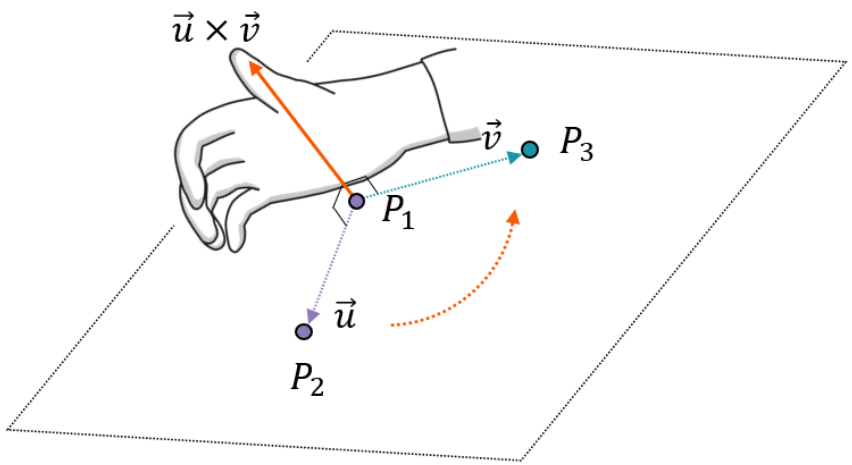
- 외적의 연산 순서에 따른 결과 비교



법선 벡터

$$\frac{\vec{u} \times \vec{v}}{|\vec{u} \times \vec{v}|}$$

- 오른손 법칙으로 외적 방향 결정



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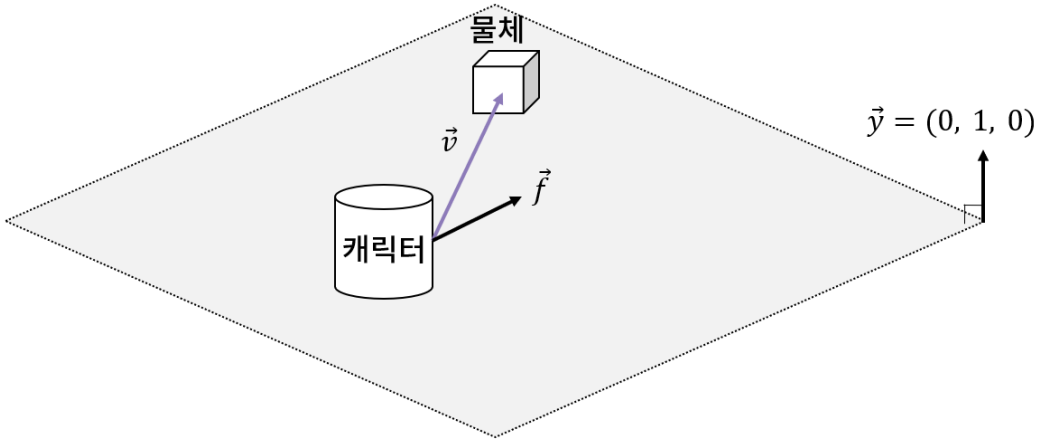
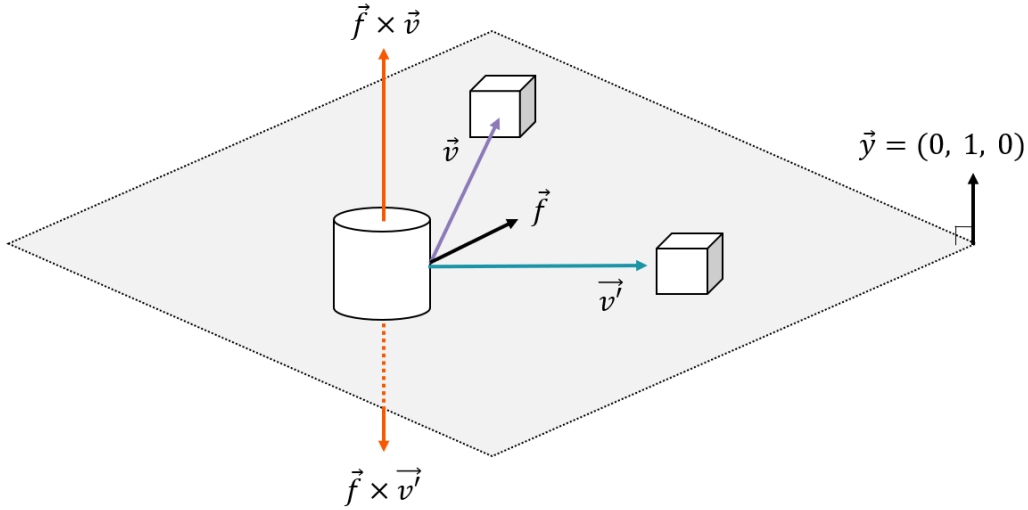
- 좌우 방향 판별을 위한 예시



- 물체 위치에 따라 달라지는 외적의 결과



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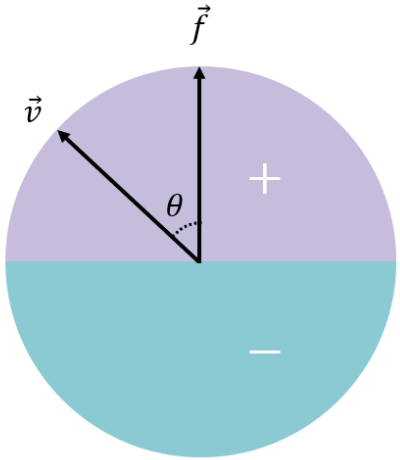
- **물체의 좌우 방향 판별을 위한 수식**

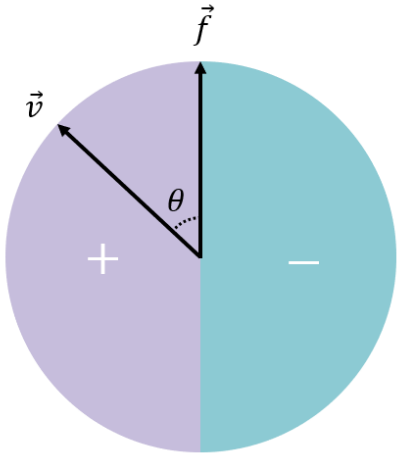
물체의 위치	판별식의 값
물체가 시선 방향에서 왼쪽에 있는 경우	$(\vec{f} \times \vec{v}) \cdot \vec{y} > 0$
물체가 시선 방향에서 오른쪽에 있는 경우	$(\vec{f} \times \vec{v}) \cdot \vec{y} < 0$
물체가 정확히 시선 방향과 일치하는 경우	$(\vec{f} \times \vec{v}) \cdot \vec{y} = 0$

- **삼각함수에 비례하는 내적과 외적이 가지는 부호 영역**

(a) 내적



(b) 외적



	내적	외적
방향 판별	앞, 뒤	왼쪽, 오른쪽

11.2 벡터로부터 회전 행렬 생성

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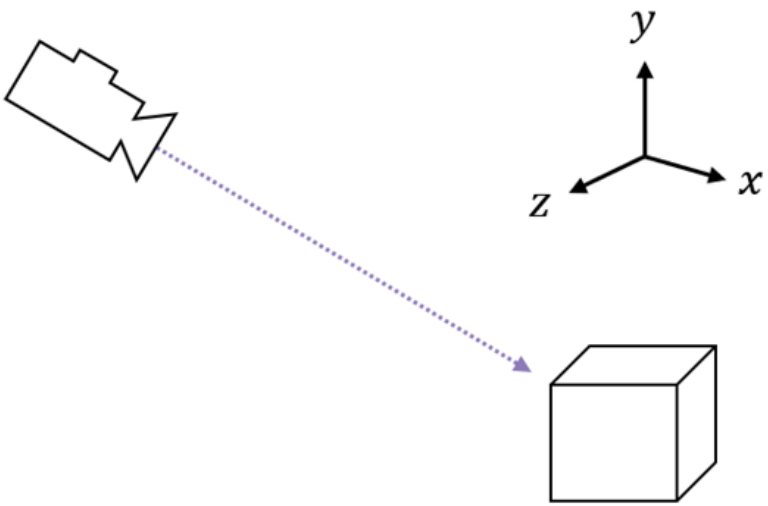
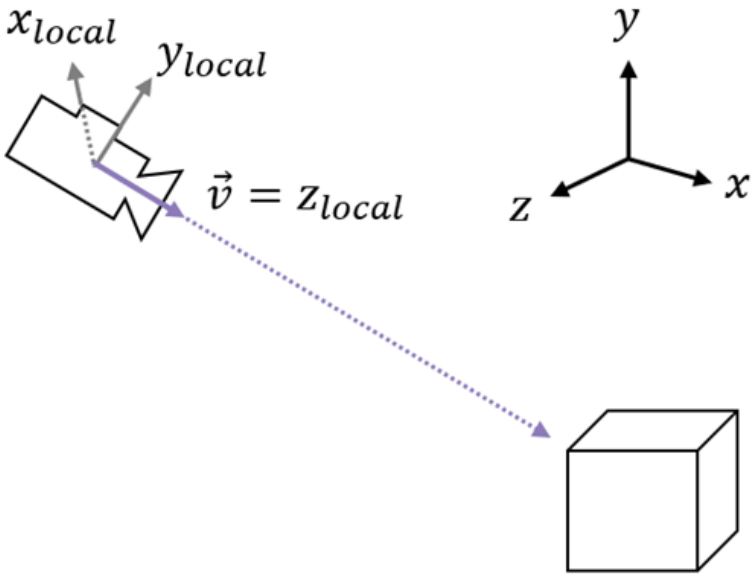
- 물체를 바라보는 카메라의 설정



- 카메라의 로컬 z축 구하기



11.2 벡터로부터 회전 행렬 생성

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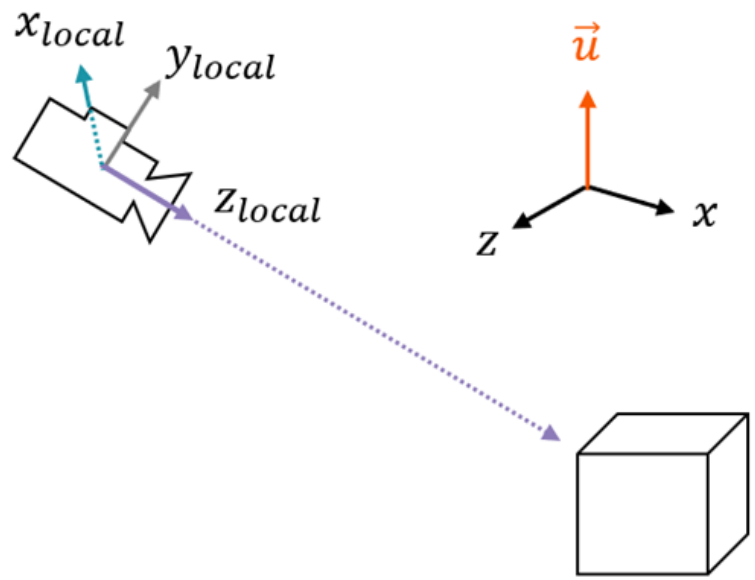
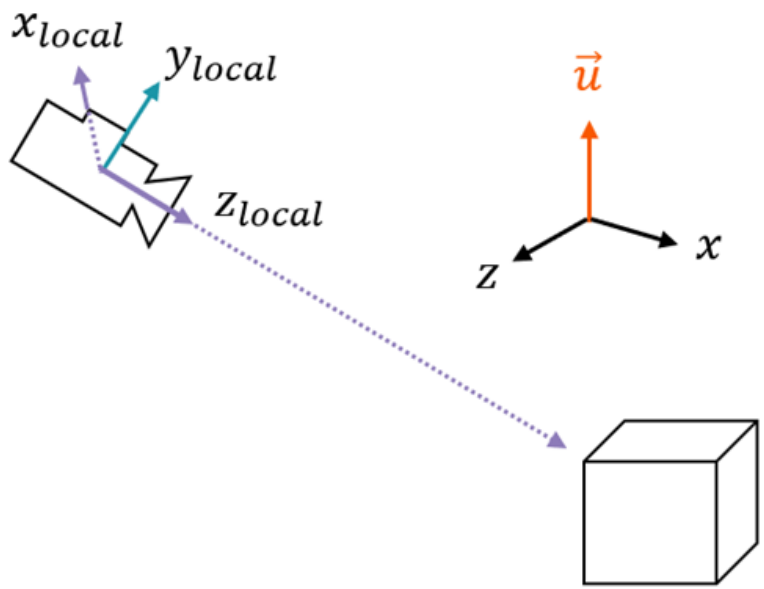
- 카메라의 로컬 x축 구하기



- 뷰 공간을 구성하는 y축 구하기



11.2 벡터로부터 회전 행렬 생성

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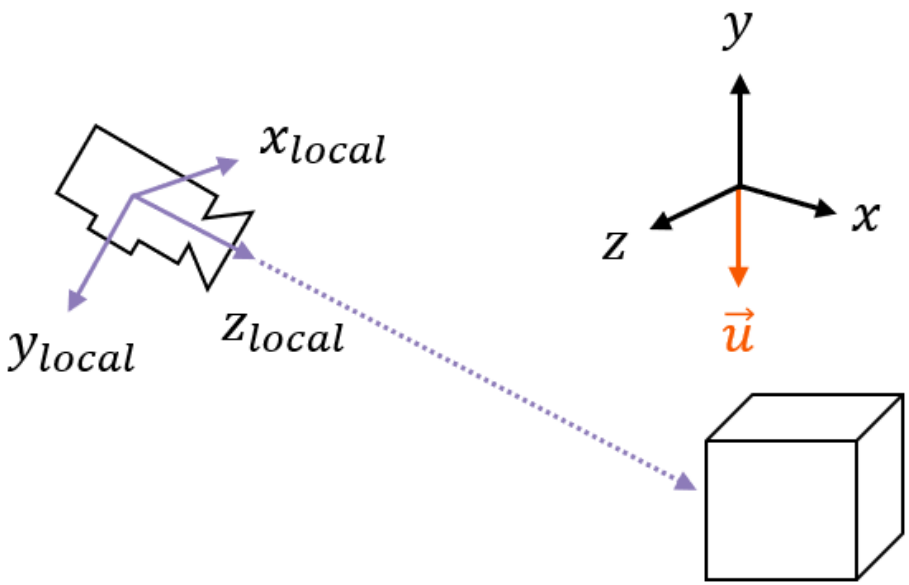
- **카메라 기저 벡터**

$$\hat{x} = \frac{\vec{u} \times \vec{v}}{|\vec{u} \times \vec{v}|} \qquad \hat{y} = \hat{z} \times \hat{x} \qquad \hat{z} = \frac{\vec{v}}{|\vec{v}|}$$

- **거꾸로 뒤집힌 카메라의 뷰 공간**

$$R = \begin{bmatrix} x_x & x_y & x_z & 0 \\ y_x & y_y & y_z & 0 \\ z_x & z_y & z_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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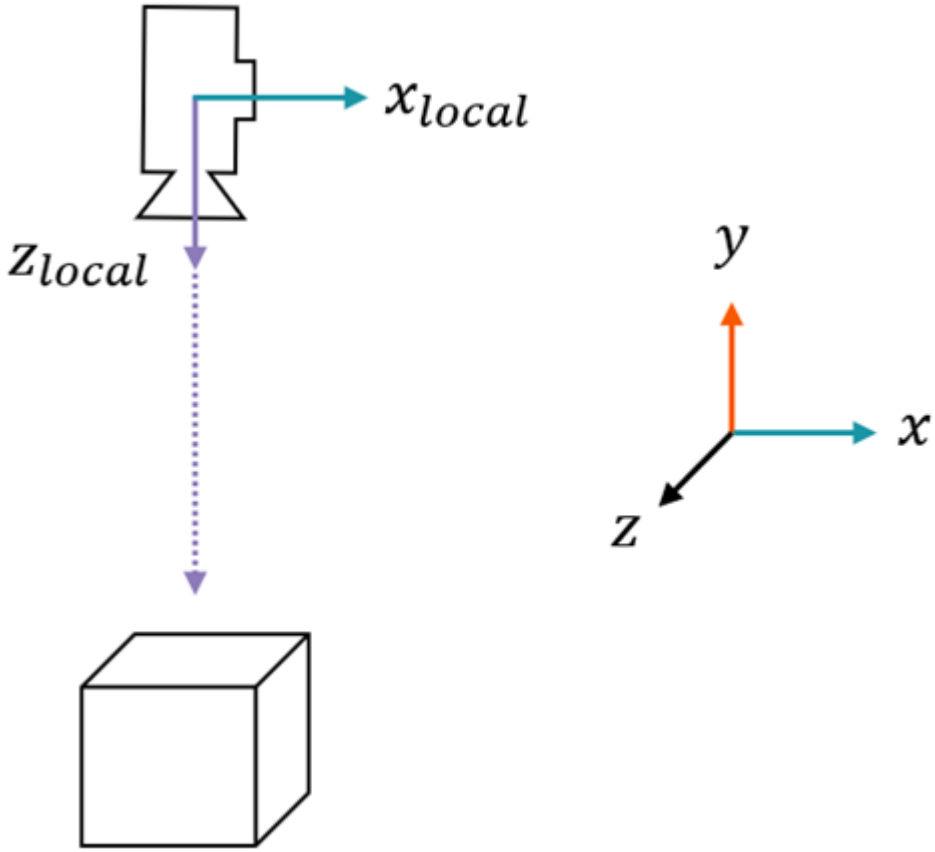
- 카메라의 시선 방향이 월드 공간 y축과 평행한 경우
  - 외적 결과 → 0-벡터
  - 외적 계산 대신 로컬 z축에 직교하는 로컬 x축의 값 수동 지정




```
// 단위 Z축 생성
localZ = (InTargetPosition
        - _Transform.GetPosition()).GetNormalize();
// 시선 방향과 월드 Y축이 평행한 경우
if (Math::Abs(localZ.Y) >= (1.f - SMALL_NUMBER))
{
  // 특이 상황에서 로컬 X 좌표 값을 임의로 지정.
  localX = Vector3::UnitX;
}
else
{
  localX = InUp.Cross(localZ).GetNormalize();
}
```

실습 11-1 항상 물체를 바라보는 카메라의 구현

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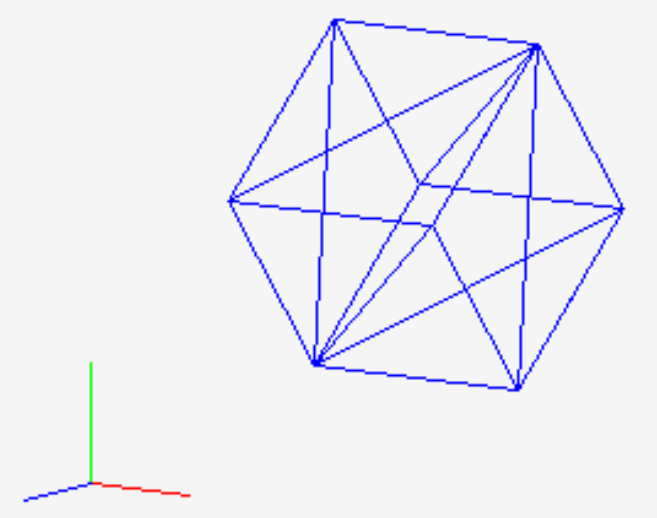

```
void CameraObject::SetLookAtRotation(const Vector3& InTargetPosition, const Vector3& InUp) {
  Vector3 localX, localY, localZ;
  // 단위 Z축 생성
  localZ = (InTargetPosition - _Transform.GetPosition()).GetNormalize();
  // 시선 방향과 월드 Y축이 평행한 경우
  if (Math::Abs(localZ.Y) >= (1.f - SMALL_NUMBER)) {
    // 특이 상황에서 로컬 X 좌표 값을 임의로 지정.
    localX = Vector3::UnitX;
  } else {
    localX = InUp.Cross(localZ).GetNormalize();
  }

  // 두 벡터가 직교하고 크기가 1이면 결과는 1을 보장
  localY = localZ.Cross(localX);
 _Transform.SetLocalAxes(localX, localY, localZ);
}
```

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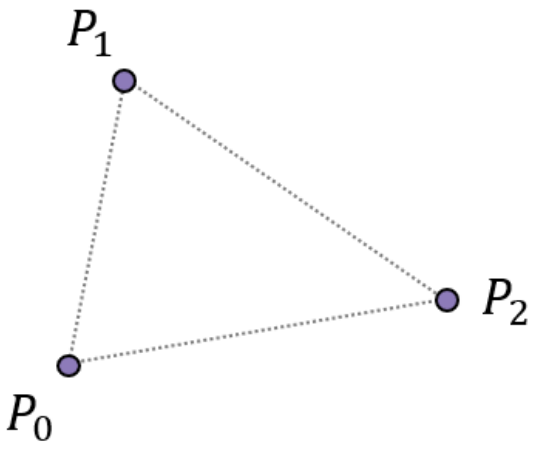
정점 버퍼와 인덱스 버퍼

정점 버퍼

인덱스
0
1
2

인덱스 버퍼

인덱스	정점 인덱스
0	0
1	1
2	2



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점의 순서에 따른 면의 방향

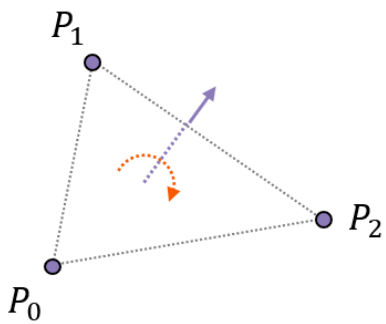
(a)

정점 버퍼

인덱스
0
1
2

인덱스 버퍼

인덱스	정점 인덱스
0	0
1	1
2	2



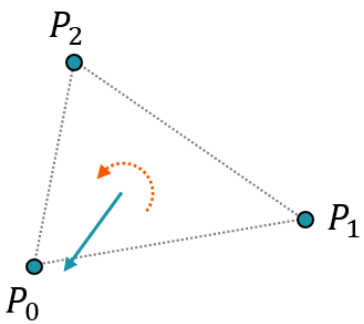
(b)

정점 버퍼

인덱스
0
1
2

인덱스 버퍼

인덱스	정점 인덱스
0	0
1	2
2	1



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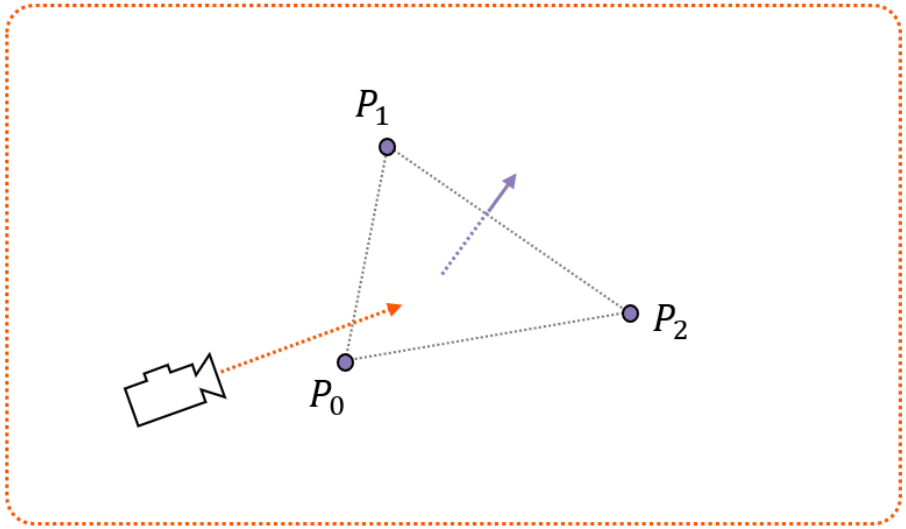
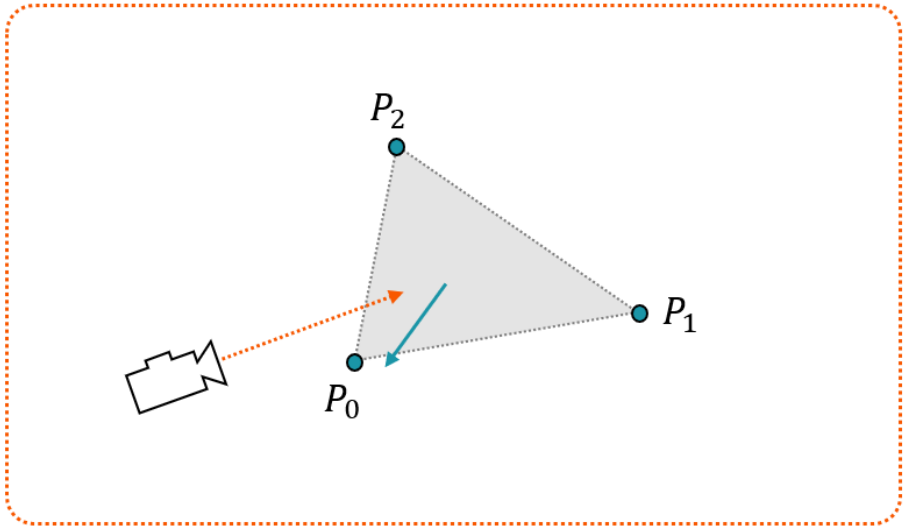
- 백페이스 컬링 원리
  - 카메라의 시선 방향, 삼각형의 법선 내적

(a)



(b)



실습11-2 보이지 않는 뒷면 그리기를 건너뛰기

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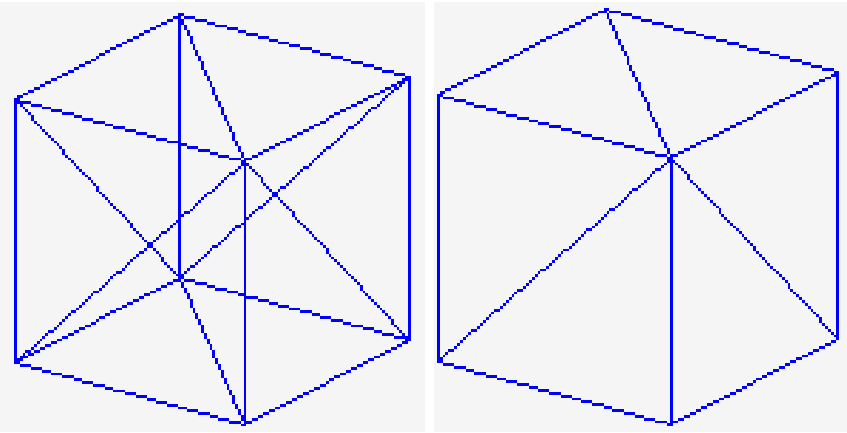
- 세 점 P_0 , P_1 , P_2

$$\vec{n} = \overrightarrow{P_0P_1} \times \overrightarrow{P_0P_2}$$

$$(n_x, n_y, n_z) \cdot (0,0,-1) = -n_z$$



```
auto& r = GetRenderer();
const GameEngine& g = Get3DGameEngine();

if (useBackfaceCulling) {
  // 백페이스 컬링 ( 뒷면이면 그리기 생략 )
  Vector3 edge1 = (InVertices[1].Position - InVertices[0].Position).ToVector3();
  Vector3 edge2 = (InVertices[2].Position - InVertices[0].Position).ToVector3();
  Vector3 faceNormal = edge1.Cross(edge2);
  Vector3 viewDirection = -Vector3::UnitZ;
  if (faceNormal.Dot(viewDirection) >= 0.f) {
    return;
```

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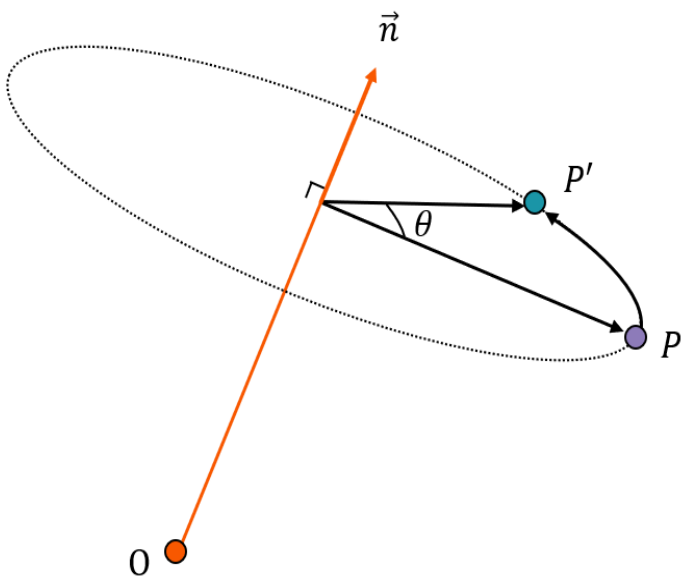
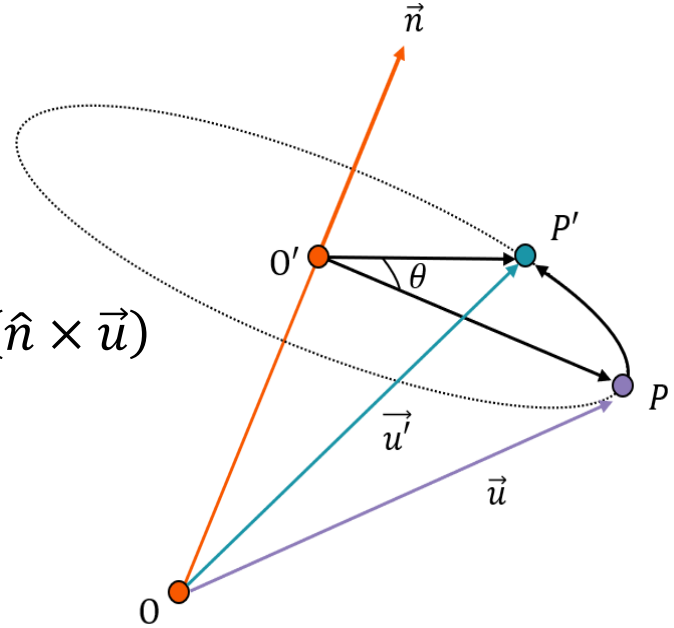
- **임의의 축에 대한 회전의 개요**



- **회전 평면의 설정**

$P = (x, y, z, 1)$ $\quad \vec{u} = P - 0 = (x, y, z, 0)$

$$\vec{u} = \cos\theta \cdot \vec{u} + (1 - \cos\theta) \cdot (\vec{u} \cdot \hat{n}) \cdot \hat{n} + \sin\theta \cdot (\hat{n} \times \vec{u})$$



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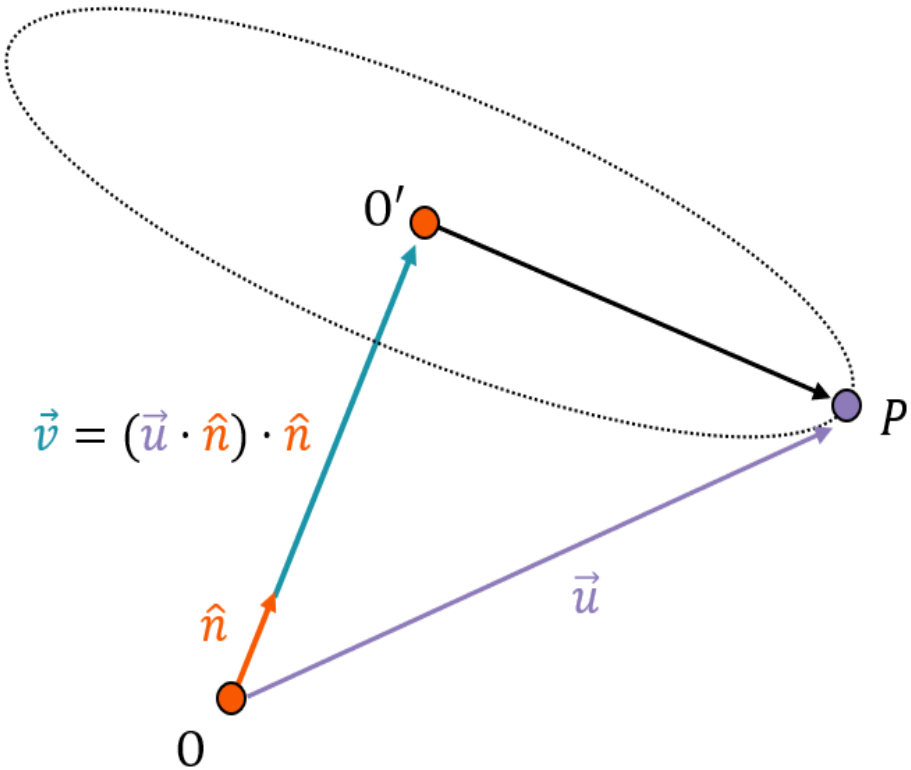
- **벡터 내적의 투영 공식으로 구한 벡터 $\vec{v}$**

$$\overrightarrow{OO'} = \vec{v} = (\vec{u} \cdot \hat{n}) \cdot \hat{n}$$



11.4 로드리게스 회전 공식

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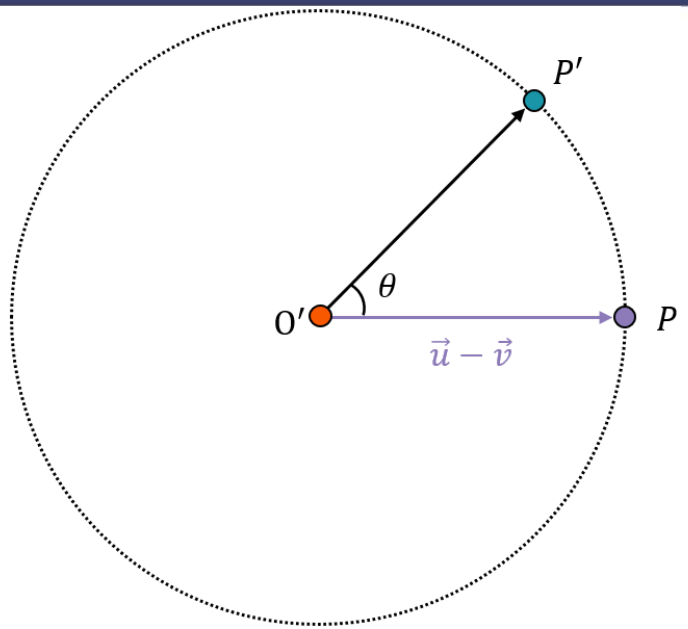
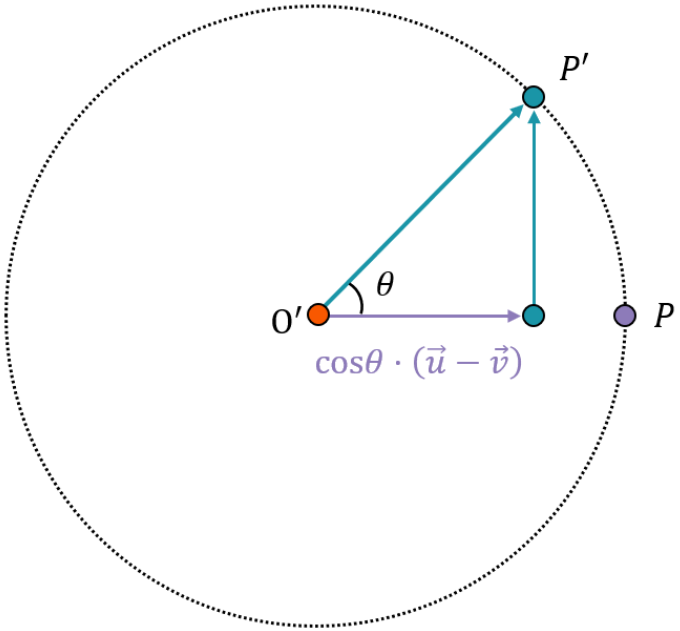
- **회전 평면에서의 회전**

$$\overrightarrow{0'P} = \vec{u} - \vec{v}$$



- **회전한 점 P'를 구하기 위한 가로 벡터의 계산**

$$\overrightarrow{0'P'} = \cos\theta \cdot \overrightarrow{0'P} + \sin\theta \cdot \overrightarrow{0'Q}$$



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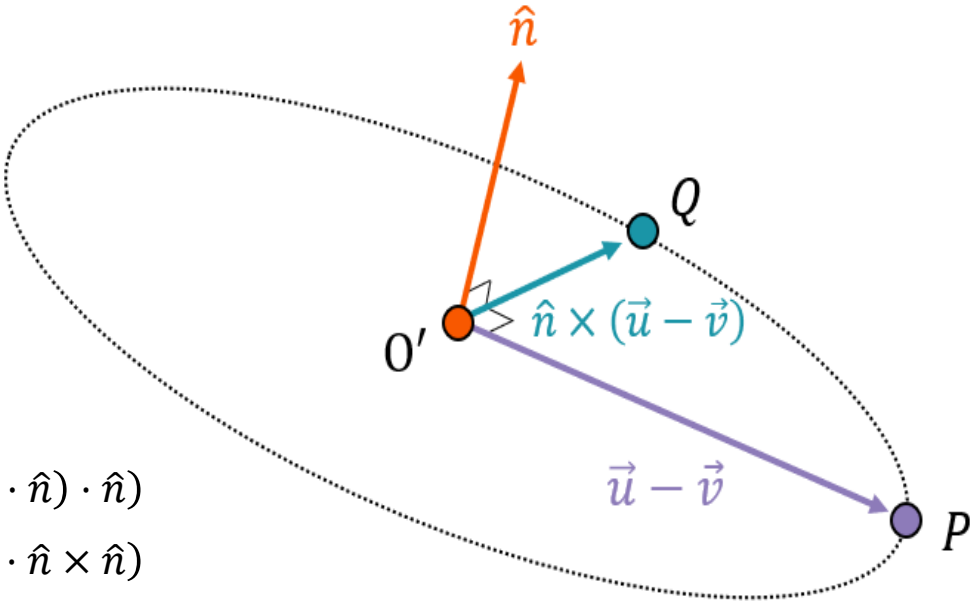
- **회전 평면에서 벡터 $\overrightarrow{O'P}$ 에 직교하는 벡터 $\overrightarrow{O'Q}$**

$$\overrightarrow{O'P'} = \cos\theta \cdot \overrightarrow{O'P} + \sin\theta \cdot \overrightarrow{O'Q}$$

$$\vec{v} = (\vec{u} \cdot \hat{n}) \cdot \hat{n}$$



$$\begin{aligned}
\overrightarrow{O'P'} &= \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot \left(\hat{n} \times (\vec{u} - \vec{v})\right) \\
&= \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot (\hat{n} \times \vec{u} - \hat{n} \times \vec{v}) \\
&= \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot (\hat{n} \times \vec{u} - \hat{n} \times (\vec{u} \cdot \hat{n}) \cdot \hat{n}) \\
&= \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot (\hat{n} \times \vec{u} - (\vec{u} \cdot \hat{n}) \cdot \hat{n} \times \hat{n}) \\
&= \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot (\hat{n} \times \vec{u})
\end{aligned}$$

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- **최종 벡터 $\overrightarrow{O'P}$의 계산**

$$\vec{v} = (\vec{u} \cdot \hat{n}) \cdot \hat{n}$$

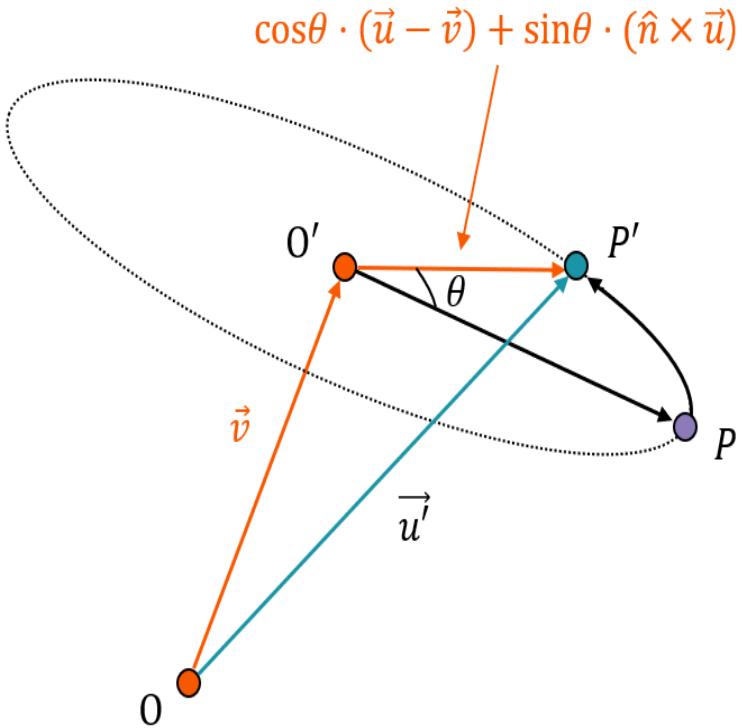
$$\overrightarrow{OP'} = \vec{v} + \overrightarrow{OP'}$$

$$\overrightarrow{O'P'} = \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot (\hat{n} \times \vec{u})$$

$$\overrightarrow{OP'} = \vec{v} + \cos\theta \cdot (\vec{u} - \vec{v}) + \sin\theta \cdot (\hat{n} \times \vec{u})$$

$$= \cos\theta \cdot \vec{u} + (1 - \cos\theta) \cdot \vec{v} + \sin\theta \cdot (\hat{n} \times \vec{u})$$

$$= \cos\theta \cdot \vec{u} + (1 - \cos\theta)(\vec{u} \cdot \hat{n}) \cdot \hat{n} + \sin\theta \cdot (\hat{n} \times \vec{u})$$



- **로드리게스 회전 공식**

$$\vec{u'} = \cos\theta \cdot \vec{u} + (1 - \cos\theta)(\vec{u} \cdot \hat{n}) \cdot \hat{n} + \sin\theta \cdot (\hat{n} \times \vec{u})$$

실습 11-3 3차원 공간에서의 자유로운 회전

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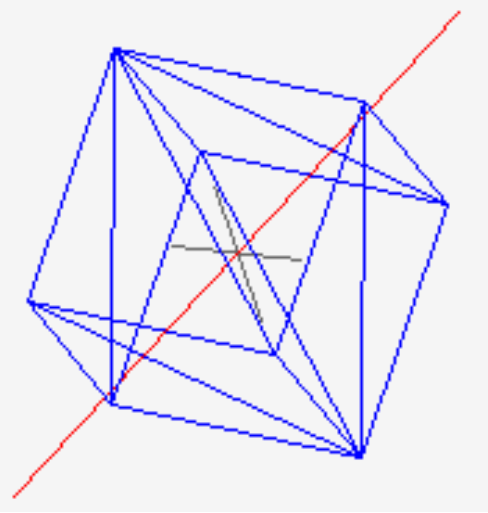
- 로드리게스 회전 공식은
  sin, cos 계산 필요
  → 행렬 변환이 용이한 사원수 사용



```
float sin = 0.f, cos = 0.f;
Math::GetSinCos(sin, cos, thetaDegree);

Vector3 u = v.Position.ToVector3();

float udotn = u.Dot(n);

Vector3 ncrossu = n.Cross(u);

Vector3 result
= Vector3(u * cos + n * ((1.f - cos) * udotn) + ncrossu * sin) * InScale;

v.Position = InMatrix * Vector4(result);
```

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- **벡터 3중 곱**

$$\vec{u} \cdot (\vec{v} \cdot \vec{w})$$

$$\vec{u} \cdot (\vec{v} \times \vec{w})$$

$$\vec{u} \times (\vec{v} \cdot \vec{w})$$

$$\vec{u} \times (\vec{v} \times \vec{w})$$

11.5.1 스칼라 삼중곱

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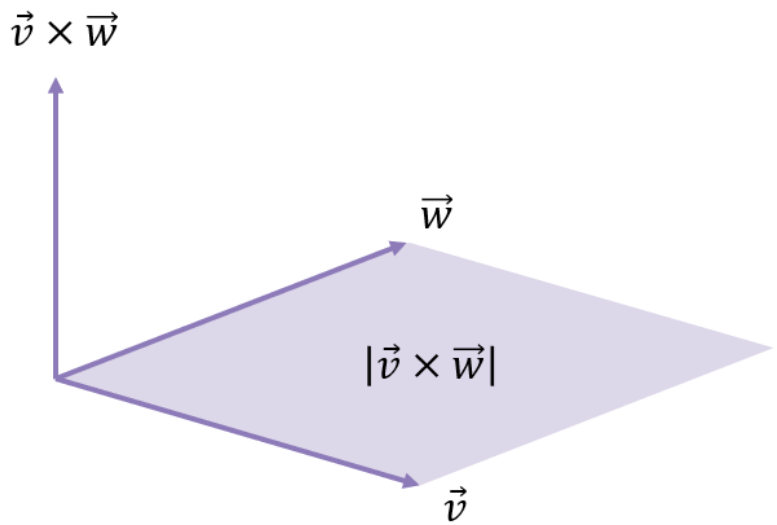
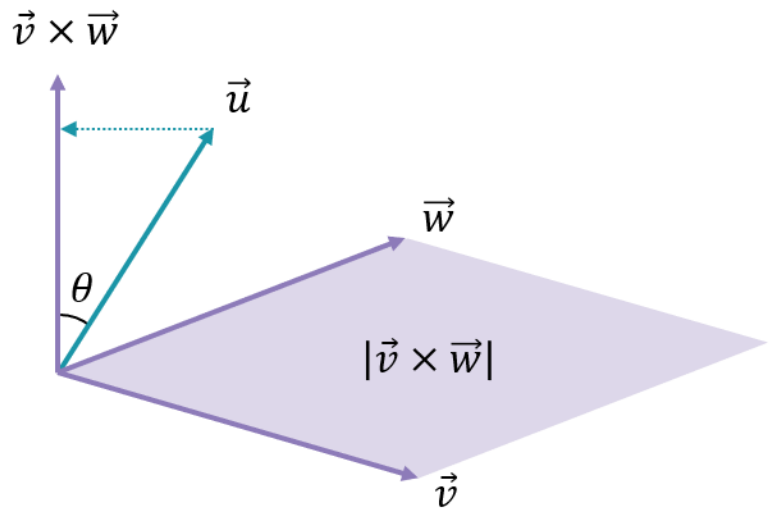
- **외적의 크기**

$$(\vec{v} \times \vec{w})$$



- **법선 벡터에 투영한 벡터의 크기**

$$(\vec{v} \times \vec{w}),\ \vec{u}$$



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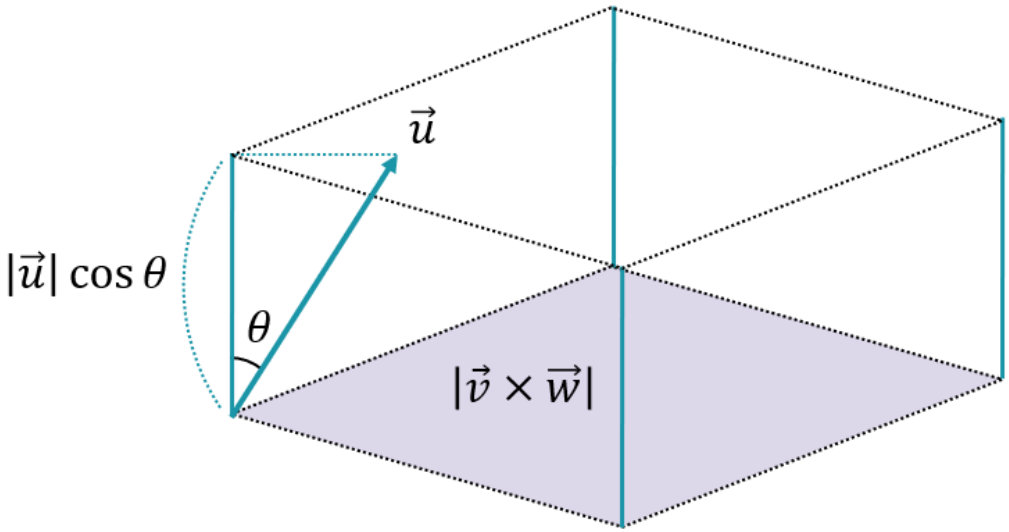
- **육면체의 부피**

$$|\vec{u}|\cos\theta$$

$$|\vec{v}\times\vec{w}|$$



11.5.1 스칼라 삼중곱

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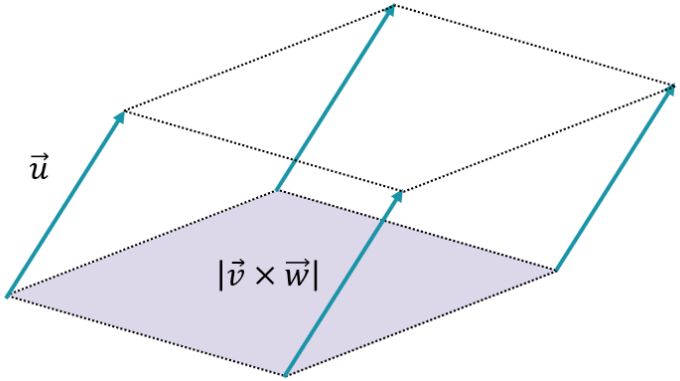
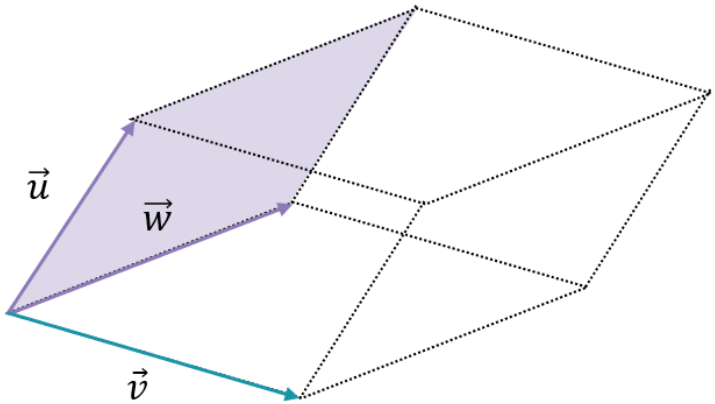
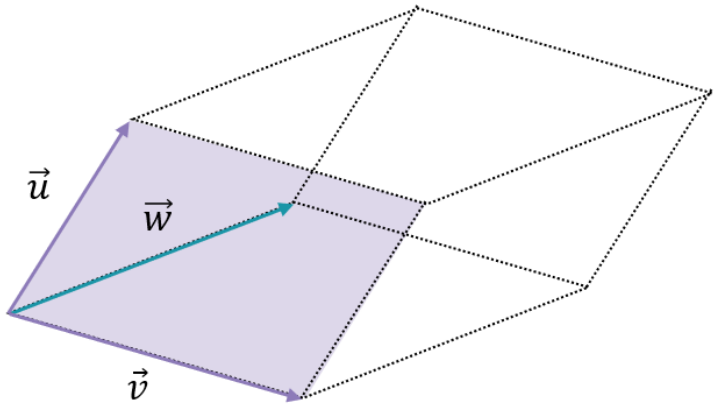
- 세 벡터로 구성된 평행육면체



- 순서를 바꿔도 결과는 동일한 스칼라 삼중곱 예시





11.5.1 스칼라 삼중곱

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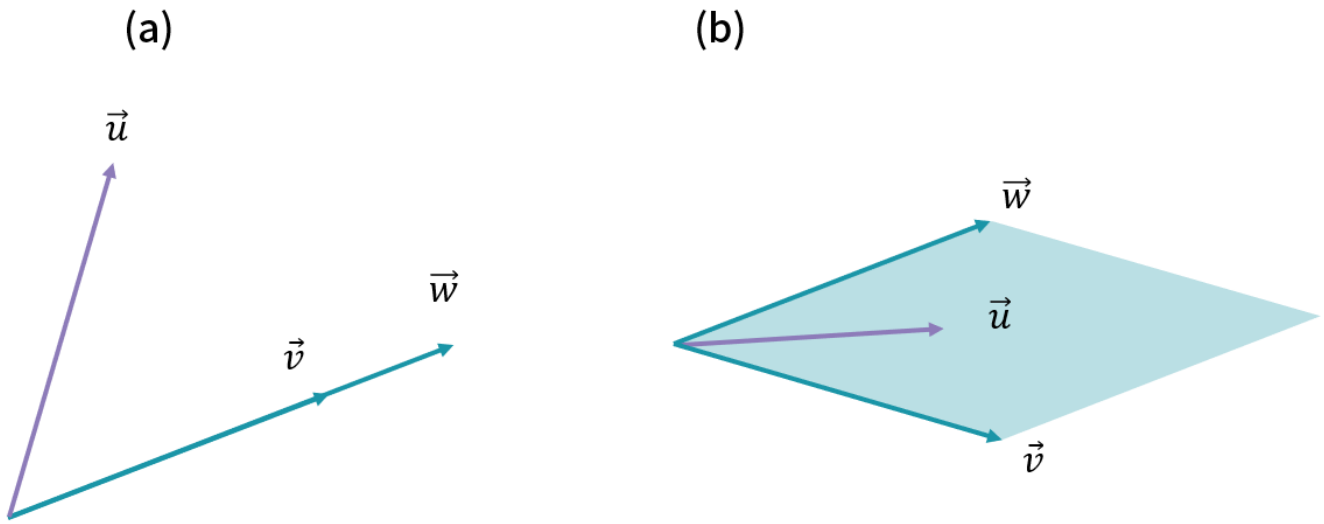
- **순서를 바꿔도 동일한 결과**

$$\vec{u}\cdot(\vec{v}\times\vec{w}) \qquad \vec{v}\cdot(\vec{w}\times\vec{u}) \qquad \vec{w}\cdot(\vec{u}\times\vec{v})$$

- **세 벡터가 모두 선형 독립이 아닌 경우**
  - **삼중곱 → 0**



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- **벡터 삼중곱 라그랑주 공식(Lagrange's formula)**

$$\left(\vec{u} \times (\vec{v} \times \vec{w})\right)_z = u_x(\vec{v} \times \vec{w})_y - u_y(\vec{v} \times \vec{w})_x$$

$$= u_x(v_z w_x - v_x w_z) - u_y\left(v_y w_z - v_z w_y\right)$$

$$= u_x v_z w_x - u_x v_x w_z - u_y v_y w_z + u_y v_z w_y$$

$$= u_x w_x v_z + u_y w_y v_z - u_x v_x w_z - u_y v_y w_z$$

$$= u_x w_x v_z + u_y w_y v_z + u_z w_z v_z - u_z w_z v_z - u_x v_x w_z - u_y v_y w_z$$

$$= \left(u_x w_x + u_y w_y + u_z w_z\right) v_z - \left(u_z v_z + u_x v_x + u_y v_y\right) w_z$$

$$= (\vec{u} \cdot \vec{w}) v_z - \left(u_x v_x + u_y v_y + u_z v_z\right) w_z$$

$$= (\vec{u} \cdot \vec{w}) v_z - (\vec{u} \cdot \vec{v}) w_z$$

$$\vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{u} \cdot \vec{v})\vec{w}$$

11.5.2 벡터 삼중곱

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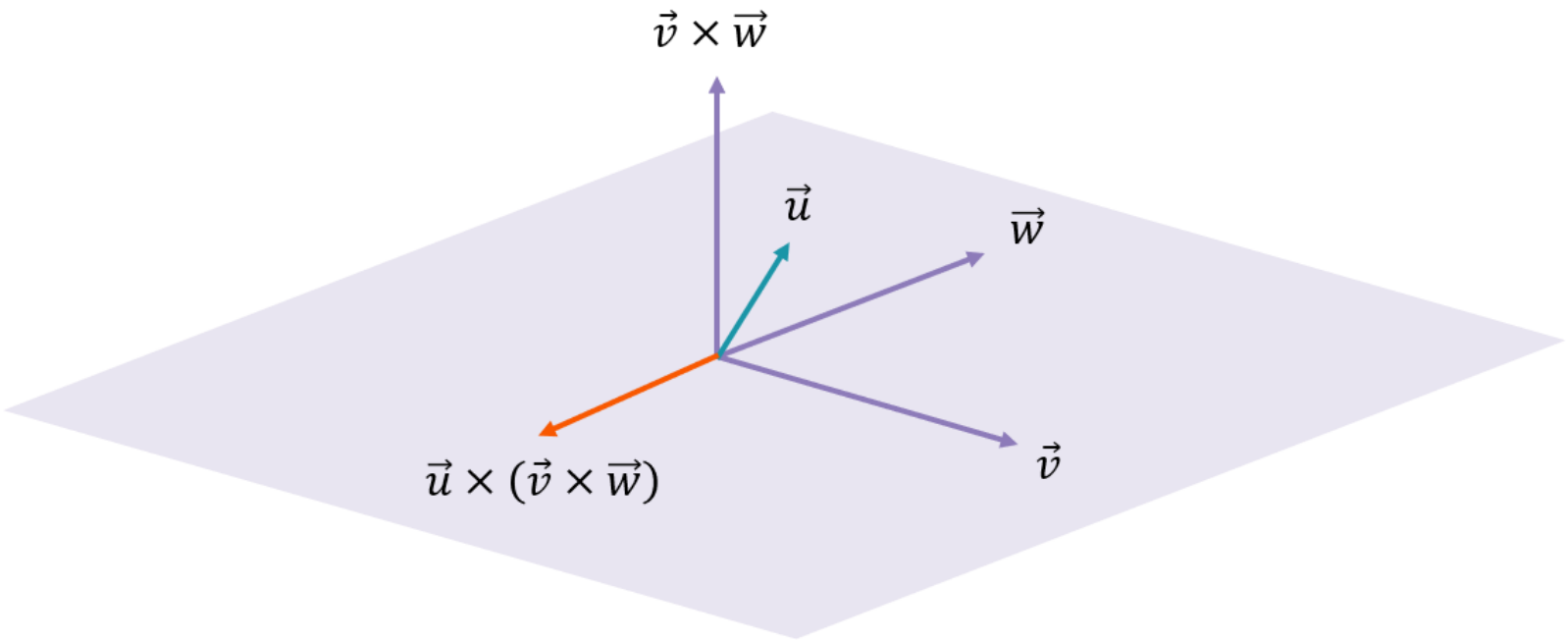
- 벡터 삼중곱으로 만들어진 벡터의 결과
  → $\vec{v}$ 와 $\vec{w}$ 가 만드는 평면의 벡터



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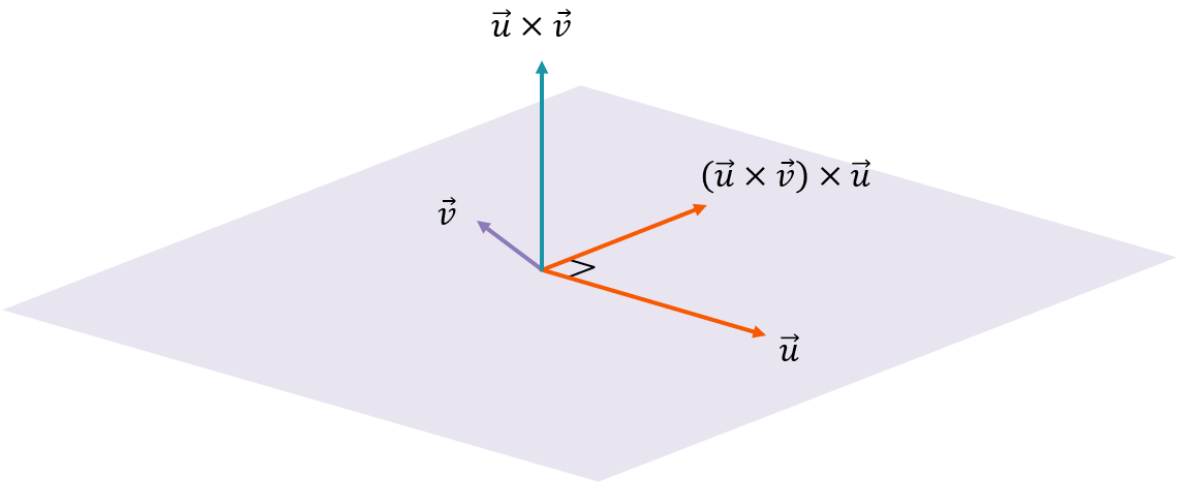
- **2차원 공간 응용**
  - 벡터 $\vec{u}$ 와 $\vec{v}$ 가 주어질 때 → u, v 평면에서 $\vec{u}$ 와 직교 벡터(예각)

$$(\vec{u} \times \vec{v}) \times \vec{u}$$



11.6 정리

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- 외적: 3차원 공간만 사용 가능
- 법선 벡터
- 로드리게스 회전 공식
- 라그랑주 공식
- 삼중곱 연산